



Talk to me...

and get a meaningful response! Meet computer systems with near human understanding

“Hi Honey, I’m home. Get me some black coffee, read out today’s headlines and, uh, just pour Garfield some milk, OK?” says Jon. No, he’s not talking to his subservient significant other. This is the future, and he’s talking to his Natural Language Understanding Enabled computer-bot named ‘Honey’.

But that is the future. Right now, Honey doesn’t exist, and your computer doesn’t understand you. The complexity of human speech and writing, and the speed at which it changes to include slang makes it difficult for a computer to understand it. To get an idea about the complexities in human language, look at these two sentences, as commands spoken to a robot.

1. “Get me some water, I’m thirsty.

While you’re about it, yeah, like, get some for Jon, too.”

2. “Procure that quantity of water that is typically consumed by a human being when he or she is thirsty. Then place that water in an appropriate container. Immediately after that, procure a similar quantity of water and place it in a different, appropriate container. Give me the first container, and give Jon the second container.”

If we all spoke and wrote in the manner of the second sentence, Natural Language Processing (NLP) might not be so hard. But we do not, cannot and will not, speak this way, right?

In the first sentence, ‘get’ is ambiguous—does it mean ‘receive’ or ‘bring’? Similarly, how much exactly is ‘some’ water? A human being would not respond to this

request by bringing, say, a bucketful of water. But it’s not so easy for the robot to figure out what ‘some’ means. ‘While you’re about it’ is an expression that needs to be programmed into the robot, because the ‘it’ is obvious only to humans. Finally, when you say “get some for Jon”, it is not obvious to the robot that you want it to get ‘some water’ for Jon too.

In Dragon Naturally Speaking, a popular NLP software, what you say to the computer gets typed out in a program, say, Microsoft Word. Sounds neat doesn’t it? But those of you who have used it know that there’s a lot to be fixed. And speech recognition is only the tip of the NLP or Natural Language Understanding (NLU) iceberg.

Speech recognition is certainly one of the most important applications of NLP,